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Exercise 2: SQL Aggregate Functions & Operators

1. The Students Table

1. General Syntax

```
SELECT DISTINCT (department)
FROM Students;
```

Results

department
IT
HR
Finance

2. General Syntax

```
SELECT department,
      AVG(age) AS avg-age
FROM Students
GROUP BY department;
```

Results

department	avg-age
IT	20.5
HR	22
Finance	23

3. General Syntax

```
SELECT COUNT (*) AS student-count,  
        department
```

```
FROM Students
```

```
GROUP BY department
```

```
HAVING COUNT (*) > 1;
```

Results department	student-count
IT	2
HR	2

4. General Syntax

```
SELECT student-id,  
        name,  
        age,  
        department
```

```
FROM Students
```

```
WHERE mark Between 21 and 23;
```

Results

Student-id	name	age	department
3	Charlie	21	IT
2	Bob	22	HR
5	Eve	22	HR
4	Diana	23	Finance

5. General Syntax

```
SELECT student-id,  
       name,  
       age,  
       department
```

```
FROM Students
```

```
WHERE dep IN ('IT', 'HR')  
      age > 21 ;
```

Results

Student-id	Name	age	department
2	Bob	22	HR
5	Eve	22	HR

Q2: The course Table

Q6. General Syntax

```
SELECT SUM (credits) AS total-credits,  
       department
```

```
FROM Students
```

```
GROUP BY department
```

```
HAVING total-credits > 5;
```

department	total-credits
IT	11
Finance	2
HR	3

Results

department	total-credits
IT	11

7. General Syntax

```
SELECT course-id,  
       Course-name,  
       department,  
       Credits
```

FROM Courses

GROUP BY Credits

WHERE Credits < > 4 ;

Course-id	Course-name	department	Credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. General Syntax

```
SELECT course-id,  
       Course-name,  
       Credits
```

FROM Courses

ORDER BY Credits DESC

Results

Course-id	Course-name	Credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

03 : The enrollments Table

9. SELECT max(grade) AS max-grade,
min(grade) AS min-grade,
AVG(grade) AS avg-grade
FROM enrollment ;

Results

max-grade	min-grade	avg-grade
90	78	84.6

10. General Syntax

SELECT COUNT(DISTINCT enrollments) AS enrollment-count,
Course-id
FROM enrollments
GROUP BY course-id ;

Course-id	enrollment-count
101	1
102	1
103	1
104	1
105	1

04 : The salaries Table

11. General Syntax

SELECT department,
Sum(salary) AS total-salary
Sum(bonus) AS total-bonus

FROM salaries

GROUP BY department ;

Results

department	total - salary	total - bonus
IT	122 000	10 500
HR	109 000	75 000
Finance	70 000	6 000

12. General Syntax

```
SELECT department
      AVG(salary) AS avg-salary
```

FROM salaries

GROUP BY department

HAVING avg-salary > 55000

Results

department	avg-salary
IT	61 000
Finance	70 000

13. General Syntax

```
SELECT employee-id,
       name,
       salary,
       bonus
```

salary + bonus AS total-compensation

FROM salaries

WHERE salary + bonus > 60000;

Results

employee-id	name	salary	bonus	total-compensation
1	Tom	60 000	5 000	65 000
3	Spike	70 000	6 000	76 000
4	Tyke	62 000	5 500	67 500

05 The projects Table

14. General Syntax

```
SELECT department,  
       SUM(budget) AS total-budget,  
       AVG(budget) AS avg-budget
```

```
FROM projects
```

```
GROUP BY department
```

```
HAVING AVG(budget) > 70000;
```

Results

department	total-budget	avg-budget
IT	270 000	13 5000
Finance	80 000	80 000

15. General Syntax

```
SELECT project-id,  
       project-name,  
       department,  
       budget
```

```
FROM projects
```

```
WHERE budget BETWEEN 50000 AND 120000
```

```
AND department <> 'Marketing';
```

Results

project-id	project-name	department	budget
1	AI App	IT	120000
2	Payroll System	Finance	80000
85	HR Portal	HR	50000